

STEAMROOT · REQUIREMENTS

SteamRoot requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2, including pressure and safety requirements
0.1	2026-09-24	Amish Chadha	Initial scaffold

SteamRoot requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3. Safety requirements (R6, R7, R9 and R10) take priority over performance requirements if they conflict.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Pasteurize soil to working depth	Soil at 70 °C (158 °F) or more for 20 to 30 min at 10 to 15 cm depth across at least 80 % of the hood footprint	Heat balance calculation; later thermocouple probes at 15 cm
R2	Treat beds at a useful rate	2 m ² /h or more at 15 cm depth; 5 m ² /h or more at 5 cm depth (weed seed bank only)	Calculation from steam output and steam per m ²
R3	Generate enough steam	30 kg/h or more of saturated steam, continuous, from one firebox	Heat transfer and combustion calculation
R4	Limit fuel use	4 kg or less of dry wood (about 16 MJ/kg, 20 % moisture) per m ² at 15 cm depth	Energy balance calculation
R5	Use biomass efficiently	65 % or more of fuel energy delivered as steam, with the economizer	Energy balance calculation; later flue temperature and fuel weighing
R6	Stay near atmospheric pressure	Normal working pressure 0.1 bar gauge (1.5 psi) or less at every point on the steam side; an open vent to atmosphere that cannot be closed or isolated	Pressure drop calculation; design review
R7	Keep stored energy low	8 L or less of water in the heated section; no closed steam drum	Volume calculation from the coil geometry
R8	Run between refills	3 h or more of steaming per feed water fill	Tank volume and steam rate
R9	Protect against overpressure	A certified steam relief valve, sized for full steam output and set at the lowest available certified pressure, 1 bar (15 psi) or less, as a second line of protection behind R6	Valve rating and capacity check
R10	Protect the operator and site	Chimney outlet 2.2 m or more above ground with a spark arrestor (mesh 6 mm or finer); fire outdoors only; steam diverted to the vent, not the hood, before the hood is lifted; hood handles 60 °C or cooler	Design review; later surface temperature check
R11	Tow with farm vehicles	Loaded mass 500 kg or less with a full tank; overall width 1.5 m or less	Mass estimate from the massing model and BOM
R12	Build within the concept budget	Parts cost \$1,500 or less	Priced BOM

■ Assumptions

- Soil to 15 cm depth: bulk density 1,300 kg/m³, gravimetric water content 20 % (dry basis), starting temperature 15 °C, heated to 75 °C so that the coldest point reaches 70 °C.

- Specific heat of dry mineral soil 0.84 kJ/(kg K), of water 4.18 kJ/(kg K).
- 70 °C for 30 minutes is widely used as a soil pasteurization target for most weed seeds and fungal pathogens. Some hard-seeded weeds and heat-tolerant viruses need higher temperatures; this is accepted.
- Pressure thresholds for boiler and pressure vessel rules vary by jurisdiction. R6 and R9 are set to be well below common low-pressure thresholds, but compliance must be confirmed locally.